

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Design of a Single Cycle RISC-V Processor

Faculty of Computer Science and Engineering | CA203 Milestone 2 Report

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Introduction & Objectives

🎯 Core Objectives

- Design a fully functional **Single-Cycle RISC-V Processor**.
- Support the complete **RV32I** instruction set (excluding FENCE).
- Ensure strict adherence to modular design principles.
- Achieve 100% compatibility with the TA-provided testbench.

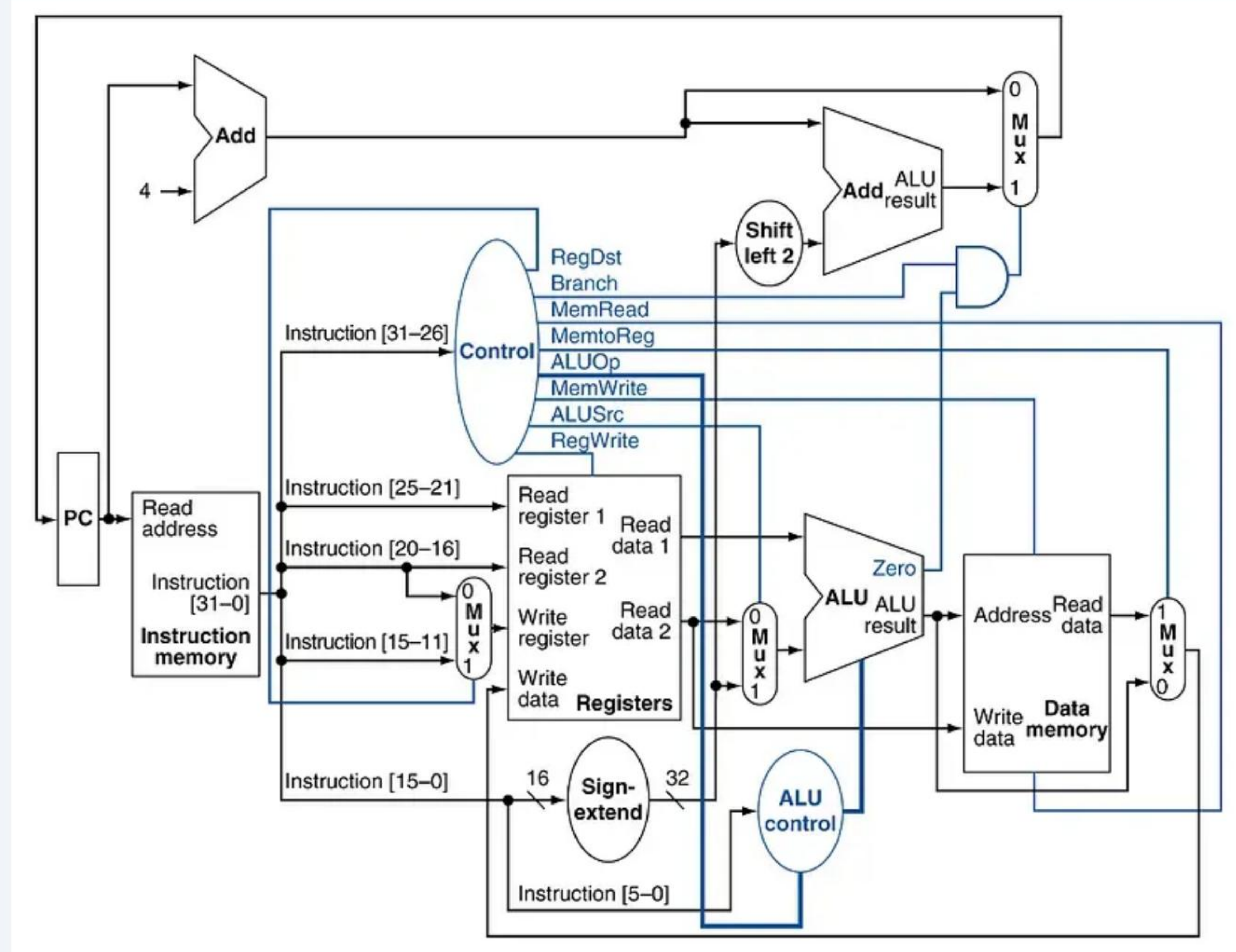
🔧 Project Scope

- **Datapath:** ALU, Register File, IMEM, DMEM, Control Unit.
- **I/O Integration:** Memory-mapped communication for LEDs, Switches, LCD, and 7-segment displays.
- **Verification:** Comprehensive simulation and testing using SystemVerilog.

Processor Datapath

The architecture follows a standard **Harvard Architecture** with separated instruction and data memory.

- ✓ **Fetch:** Retrieves instruction from 16KB IMEM.
- ✓ **Decode:** Control Unit parses Opcode/Funct3/Funct7.
- ✓ **Execute:** ALU performs arithmetic/logic ops.
- ✓ **Memory:** Accesses DMEM or I/O peripherals.
- ✓ **Writeback:** Updates the Register File.



Memory Specifications



Instruction Memory

Size: 16KB (Read-Only)

Init: Loaded from isa.mem

Access: Asynchronous Read

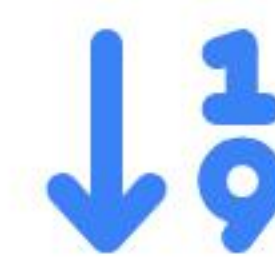


Data Memory

Size: 2KB (0x000-0x7FF)

Type: Read-Write RAM

Access: Sync Write / Async Read



Architecture

Endianness: Little-Endian

Alignment: Handles Byte, Half-word, and Word access

Addressing: Byte-addressable

Memory-Mapped I/O Map

PERIPHERAL	ADDRESS RANGE	DESCRIPTION / MAPPING
Switches (SW)	0x1001_0000 - 0x1001_0FFF	Read-only input (32 switches)
LCD Control	0x1000_4000 - 0x1000_4FFF	Control registers for LCD display
HEX Displays	0x1000_2000 - 0x1000_3FFF	HEX0-3 and HEX4-7 Control
Green LEDs	0x1000_1000 - 0x1000_1FFF	8 Green LEDs output
Red LEDs	0x1000_0000 - 0x1000_0FFF	17 Red LEDs output

Core Components Implementation

Arithmetic Logic Unit

The heart of execution, supporting 10 distinct operations defined by the ISA.

- **Arithmetic:** ADD, SUB
- **Logical:** AND, OR, XOR
- **Shifts:** SLL, SRL, SRA (Arithmetic right)
- **Comparison:** SLT (Signed), SLTU (Unsigned)

Control Unit

Decodes the 7-bit opcode to generate critical control signals.

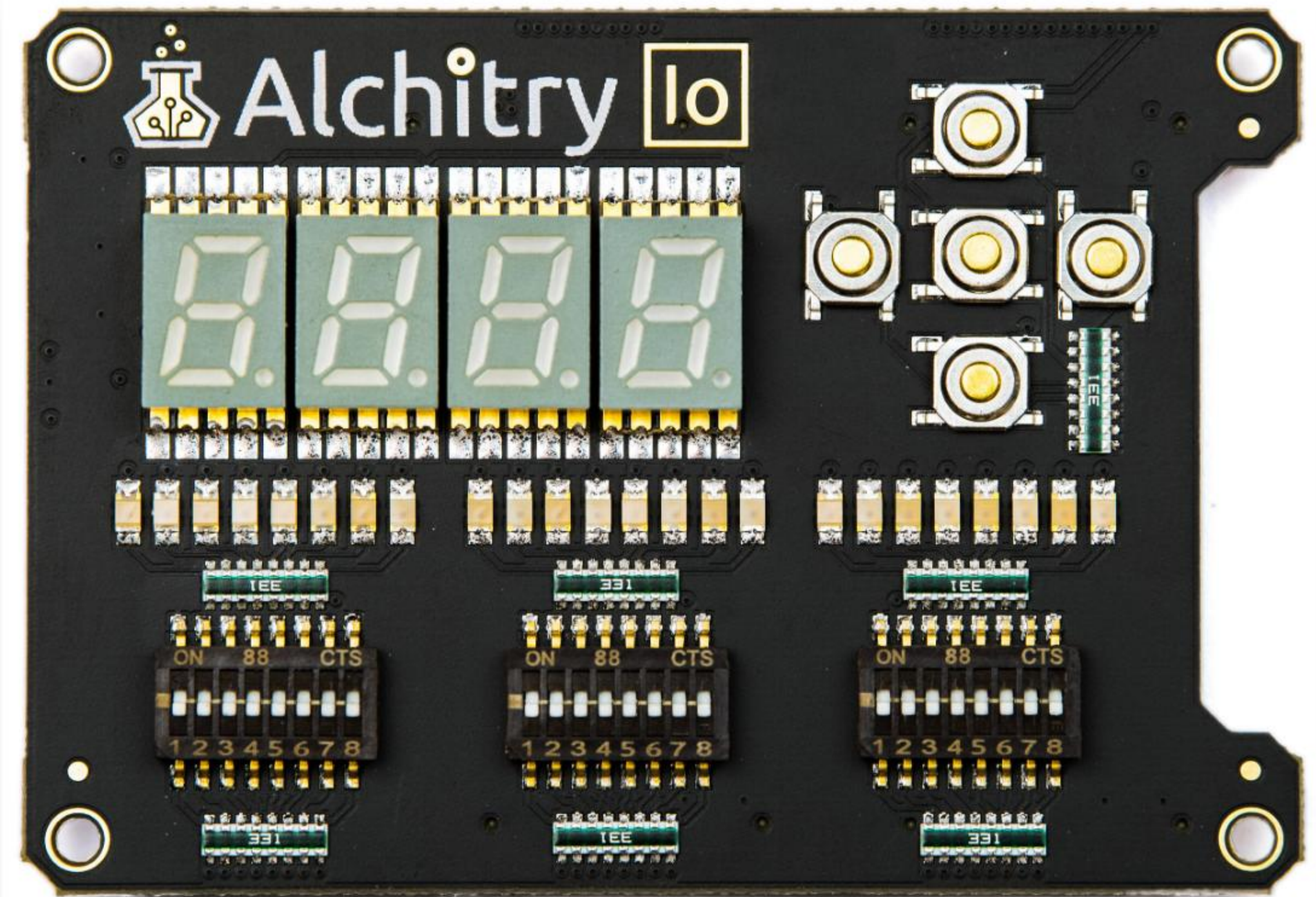
- **Signals:** reg_write, mem_read/write, alu_op, branch, jump.
- **Types Handled:** R-Type, I-Type, S-Type, B-Type, U-Type, J-Type.
- **Logic:** Purely combinational logic for single-cycle execution.

Peripheral Integration

Load-Store Unit (LSU)

The LSU is the gatekeeper for all data transactions. It distinguishes between standard memory access and I/O requests.

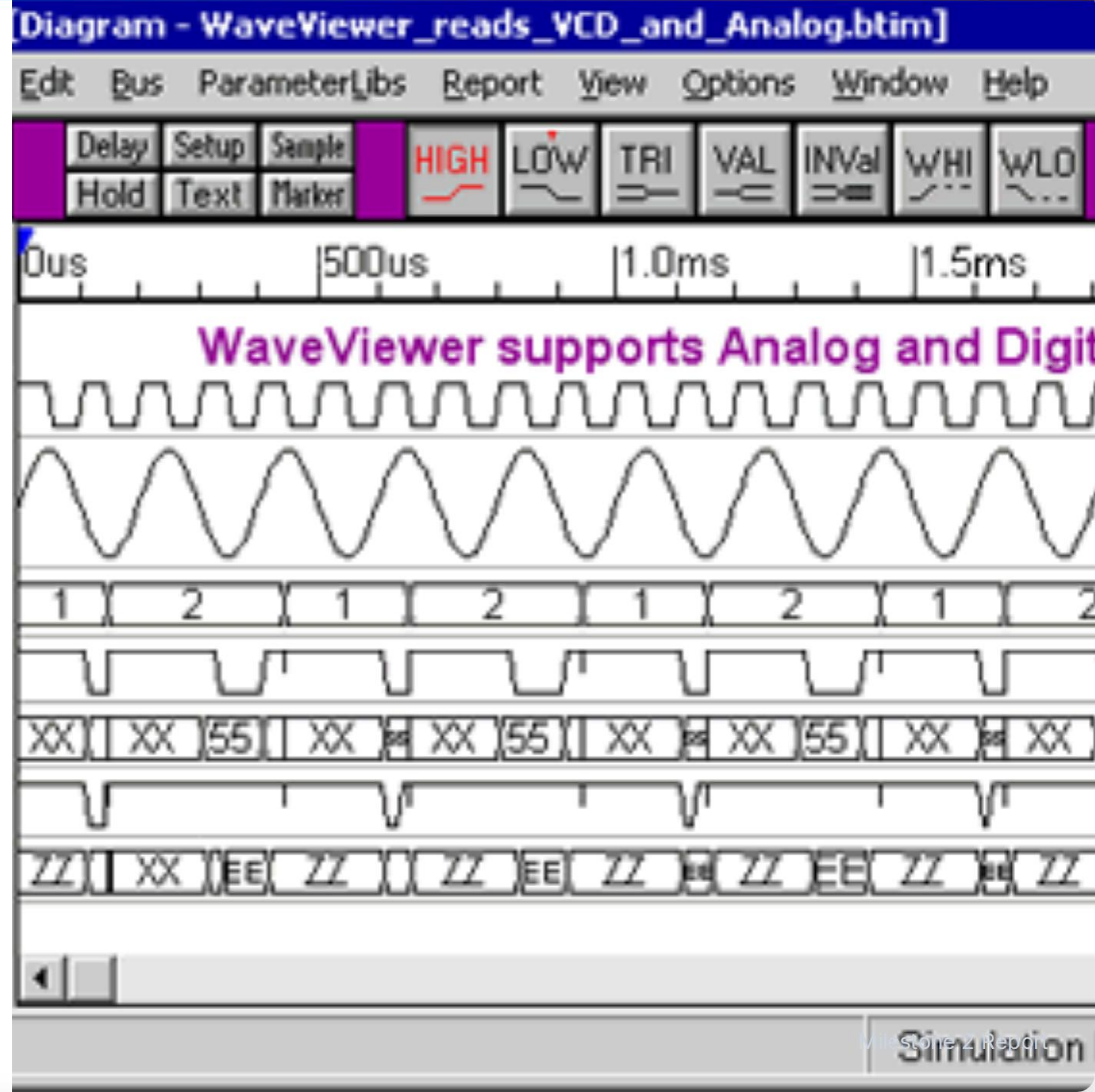
- **Address Decoding:** Routes data to DMEM (lower addresses) or I/O registers (higher addresses).
- **Write Logic:** Synchronous updates to LED/HEX registers based on mem_write.
- **Read Logic:** Muxes data from Switches or Memory back to the CPU.



Testing & Verification

We employed a rigorous verification strategy to ensure ISA compliance.

- ✓ **Methodology:** Modular testing followed by full system integration.
- ✓ **ISA Test Suite:** TA-provided isa.mem containing tests for all instruction types.
- ✓ **Simulation:** Verified using Verilator and Xcelium.
- ✓ **Waveform Analysis:** Confirmed signal integrity and correct state transitions.



Results & Achievements

41/41

Tests Passed

Full ISA Compliance Verified

100%

I/O Functional

All Peripherals Mapped Correctly

1

CPI

Single Cycle Execution Achieved

Status: The processor successfully executes all arithmetic, logical, memory, branch, and jump instructions required for Milestone 2.

Future Improvements

►► Performance

- **Pipelining:** Implement a 5-stage pipeline to increase clock frequency and throughput.
- **Branch Prediction:** Add logic to minimize control hazards and stall cycles.

≡ Advanced Features

- **Cache Memory:** Integrate Instruction and Data Caches to reduce memory access latency.
- **Interrupts & Exceptions:** Implement precise exception handling for robust operation.

"The implementation serves as a robust foundation for understanding advanced processor architectures."

— Milestone 2 Conclusion

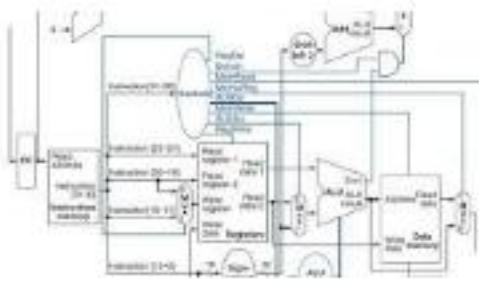
We have successfully designed, implemented, and verified a complete Single-Cycle RISC-V Processor.

Q & A

Thank you for your attention.

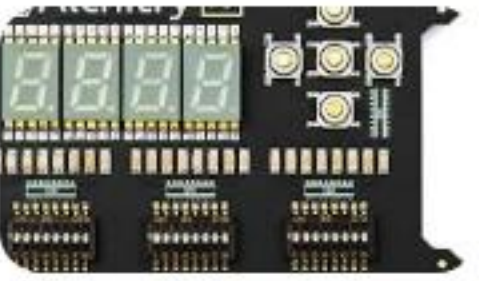
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Image Sources



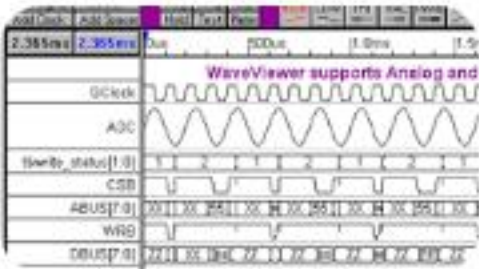
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